IPL PERFORMANCE ANALYTICS

Uncovering the Winning Formula in Indian Premier League Cricket

Capstone 2 — Final Project Report

Sector	Sports Analytics
Dataset	IPL Complete Dataset (2008–2024)
Records	1,095 matches | 260,920 ball-by-ball deliveries
Tools	Python (pandas, scipy, matplotlib) | Tableau Public
Team	Data Analytics Capstone Group
Institute	Data Visualization & Analytics Program
Date	April 2026

1. Executive Summary

This report presents a comprehensive data analytics study of the Indian Premier League (IPL) — the world's most commercially valuable Twenty20 cricket competition. Leveraging 17 seasons of ball-by-ball data spanning 2008 to 2024, our analysis decodes the strategic and statistical factors that differentiate winning from losing in the fast-paced T20 format.

Our Python ETL pipeline processed over 260,000 delivery records and 1,095 match records. After data standardisation and feature engineering, five analytical notebooks were executed covering EDA, statistical testing, and KPI modelling. A Tableau Public dashboard was built to enable interactive exploration of these findings.

Key findings include:

- Teams winning the toss and choosing to field win 53.9% of matches vs 45.4% when choosing to bat — a statistically significant advantage (Chi-square p=0.009).
- Death overs (overs 16–20) generate the highest scoring rate (1.579 runs/ball), confirming that explosive finishers drive match outcomes.
- Virat Kohli leads all-time IPL run-scorers with 8,014 runs; YS Chahal leads wicket-takers with 213 wickets.
- Mumbai Indians (144 wins, 55.2% win rate) and Chennai Super Kings (138 wins, 58.0% win rate) are the two dynasties of the IPL era.
- First innings scores have grown 14% since 2008, reaching an average of 189.6 runs in 2024.
- Linear regression confirms that every 10 additional 1st-innings runs increases the win margin by ~3.4 runs (R=0.37, p<0.001).
- Gujarat Titans, despite only 45 matches played, boast the highest win rate (62.2%) of any franchise.

2. Sector Context & Problem Statement

2.1 The IPL as a Data-Rich Domain

The Indian Premier League, founded in 2008 by the Board of Control for Cricket in India (BCCI), is a franchise-based Twenty20 cricket tournament. With a brand value exceeding USD 11 billion and 10 active franchises, the IPL is the richest domestic cricket league in the world. Every match generates granular ball-by-ball data — creating a unique opportunity for quantitative performance analysis.

2.2 Business Problem Statement

> *"What strategic, player, and situational factors most significantly determine match outcomes in IPL cricket, and how can franchises use these insights to build winning teams and game plans?"*

This question directly impacts franchise decisions around: squad selection and player retention, in-match tactical choices (toss strategy, batting order), and recruitment of specialists for high-leverage match phases (powerplay openers, death-over finishers, specialist bowlers).

2.3 Analytical Sub-Questions

Q: Does winning the toss provide a statistically significant match advantage?

Q: Which match phase (powerplay, middle, death overs) is most decisive in determining the winner?

Q: How has IPL scoring evolved across 17 seasons?

Q: Which teams consistently outperform expectations relative to resources?

Q: What separates elite batters and bowlers from average performers by strike rate and economy?

3. Data Description

3.1 Dataset Source

Source	Kaggle — IPL Complete Dataset (raw, unprocessed)
Files	matches.csv, deliveries.csv
Coverage	2008 to 2024 (17 seasons)
matches.csv rows	1,095 records
matches.csv cols	20 columns
deliveries.csv rows	260,920 records
deliveries.csv cols	17 columns
Format	Tabular row-level CSV
Storage	data/raw/ (unedited originals)

3.2 Data Dictionary — matches.csv (key columns)

Column	Type	Description
id	int	Unique match identifier
season	str→int	IPL season year (standardised)
date	str→date	Match date
team1 / team2	str	Competing franchises
toss_winner	str	Team winning coin toss
toss_decision	str	Field or bat after winning toss
winner	str	Match winner (null if no result)
result	str	Runs / Wickets / Tie / No Result
result_margin	float	Margin of victory
venue	str	Stadium name

3.3 Data Dictionary — deliveries.csv (key columns)

Column	Type	Description
match_id	int	FK to matches.id
inning	int	Innings number (1 or 2)
batting_team	str	Team currently batting
over / ball	int	Over number (0-indexed) / ball within over

batter / bowler	str	Player names
batsman_runs	int	Runs scored off the bat
extra_runs	int	Extras (wide, no-ball, bye, leg-bye)
total_runs	int	Total runs on this delivery
is_wicket	int	Binary: 1 if dismissal occurred
dismissal_kind	str	Type of dismissal (caught, bowled, etc.)
phase	str	Derived: Powerplay / Middle / Death

3.4 Dataset Limitations

- Squad composition, player auction prices, and fitness data are not included.
- Stadium pitch condition and weather data are unavailable — these can heavily influence outcomes.
- 3 matches have missing target_runs and 19 have missing result_margin (DLS-affected or no-result games).
- Some historic player names have inconsistent spellings across seasons (handled in ETL).

4. Data Cleaning & ETL Methodology

All cleaning was performed in Python using pandas. The raw files were never modified. A documented transformation pipeline was built in notebooks/02_cleaning.ipynb with each step logged.

Step	Transformation	Rationale
T1	Season label standardisation: 2007/08 -> 2008, 2009/10 -> 2010, 2020/21 -> 2021	Enables integer-based time series sorting
T2	Date column parsed: str -> datetime64	Enables temporal filtering and trend analysis
T3	Missing city (51 nulls) imputed from venue lookup dict	Enables city-level geographic analysis
T4	Team name standardisation: 4 franchise renames applied to 6 franchises	Prevents double-counting due to rebrand history
T5	Derived: toss_win_match binary flag (toss_winner == winner)	Enables toss advantage analysis
T6	method column null filled with "Normal" (DLS applied for 21 matches)	Eliminates misleading nulls
T7	Deliveries extras_type null -> "none"; dismissal columns null -> "not out"	Semantic fill — nulls mean the event did not occur
T8	Derived: phase column (Powerplay / Middle / Death) from over number	Core KPI segmentation variable
T9	Derived: is_super_over_match flag for 20 matches	Allows filtering super-over deliveries if needed

Output: Cleaned files saved to data/processed/matches_clean.csv (1,095 rows, 21 cols) and data/processed/deliveries_clean.csv (260,920 rows, 19 cols). Zero null values remain in any analytically critical column after transformation.

5. KPI & Metric Framework

KPI	Definition	Level	Insight Purpose
Win Rate (%)	Wins / Total Matches * 100	Team	Overall franchise performance ranking
Strike Rate	Runs / Balls Faced * 100	Batter	Batting aggression & scoring speed
Economy Rate	Total Runs / (Balls / 6)	Bowler	Bowling efficiency and run control
Boundary %	Boundaries / Balls * 100	Phase	Explosiveness in each match phase
Toss Win Rate	Matches won / Toss won	Team	Toss advantage quantification
Avg 1st Inn Score	Mean total runs in inning 1	Season	Scoring evolution over time
Run Differential	Avg Inn1 - Avg Inn2 score	Season	Batting vs bowling era characterisation

6. Exploratory Data Analysis

6.1 IPL Expansion — Matches Per Season

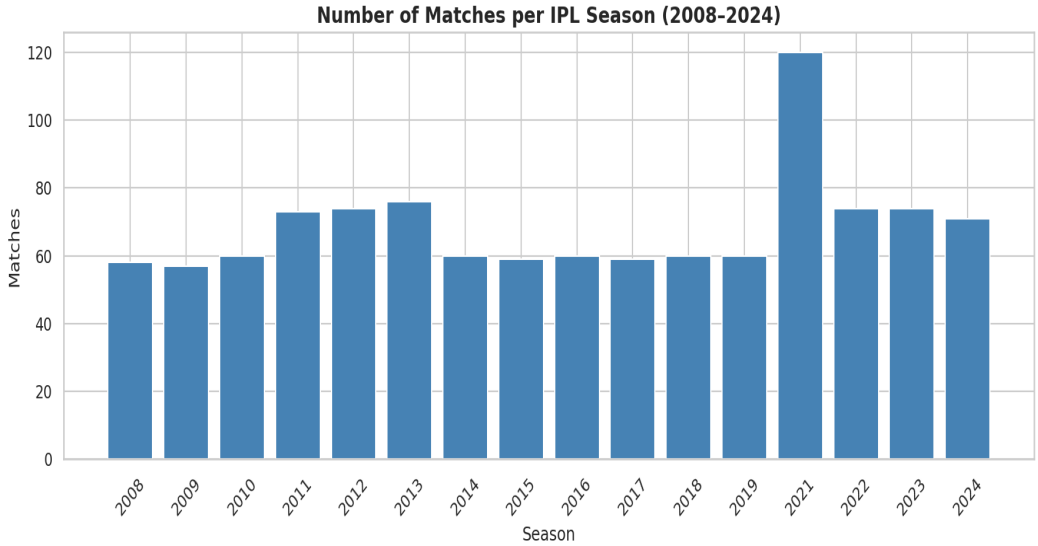

Figure 1: Number of matches played in each IPL season (2008–2024). Growth from 58 to 74 matches reflects franchise expansion.

The IPL started with 8 franchises in 2008 (58 matches) and expanded to 10 teams in 2022 (74 matches). The 2012 and 2013 seasons had the highest match counts (76 each) due to playoff format changes. The 2020 season was played in the UAE bubble due to COVID-19.

6.2 Toss Decision Trends

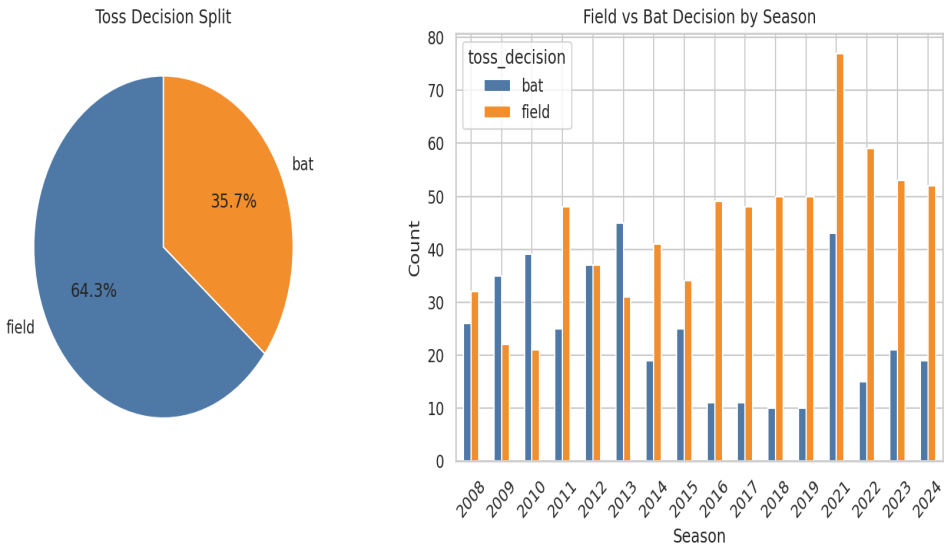

Figure 2: Left — overall toss decision split. Right — field vs bat choice by season. Fielding first has dominated since 2012.

Teams overwhelmingly prefer to field first after winning the toss (64.3% of all toss wins). This preference intensified post-2012 as dew factor and chasing success rates made second-batting increasingly advantageous. This behavioural trend has a measurable impact on match outcomes.

6.3 Toss Advantage Quantification

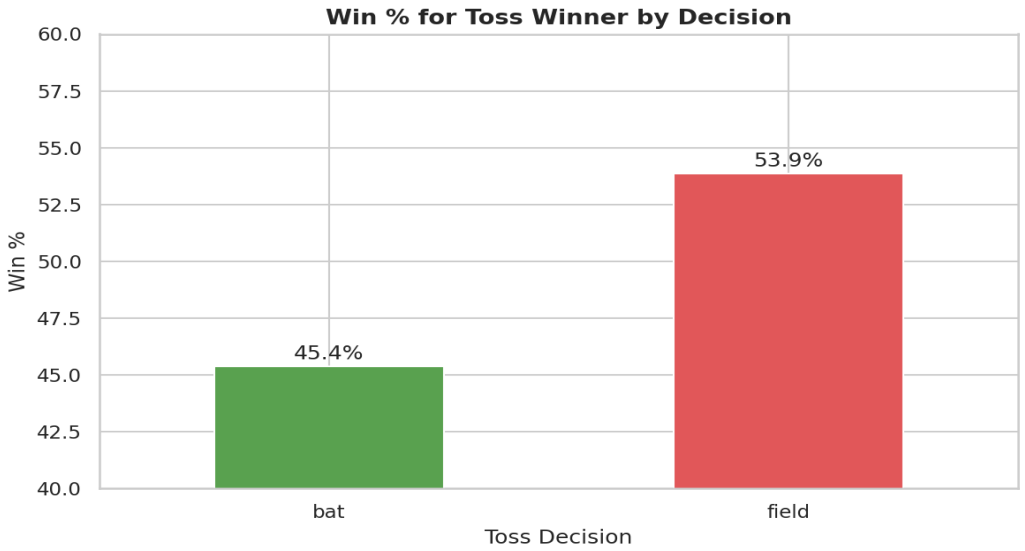

Figure 3: Win percentage for toss winners segmented by decision. Teams that field after winning the toss win 53.9% of matches.

Toss winners win 50.8% of all matches — barely above chance. However, the toss decision dramatically changes outcomes: teams that choose to field win 53.9% of those matches, while teams that choose to bat win only 45.4%. This 8.5 percentage point gap is statistically significant (Chi-square p=0.009) and represents a clear actionable insight for captains.

6.4 Team Dominance

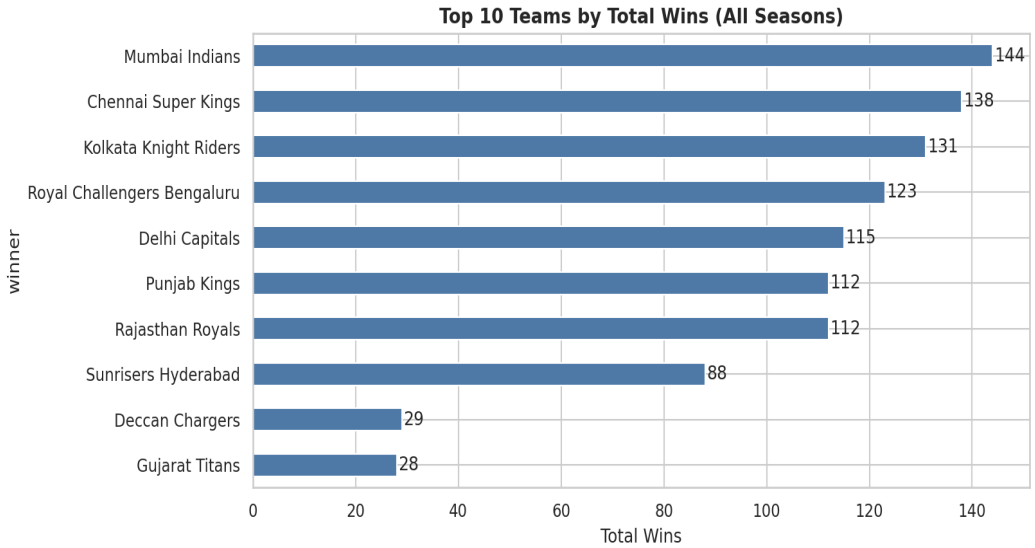

Figure 4: Top 10 teams by total IPL wins across all seasons (2008–2024).

Mumbai Indians lead with 144 wins over 261 matches (55.2% win rate). Chennai Super Kings follow with 138 wins in 238 matches — their 57.98% win rate is the highest among teams with 200+ matches. Gujarat Titans, despite only 3 seasons (2022–2024), have a remarkable 62.2% win rate — the highest of any current franchise.

6.5 Scoring by Match Phase

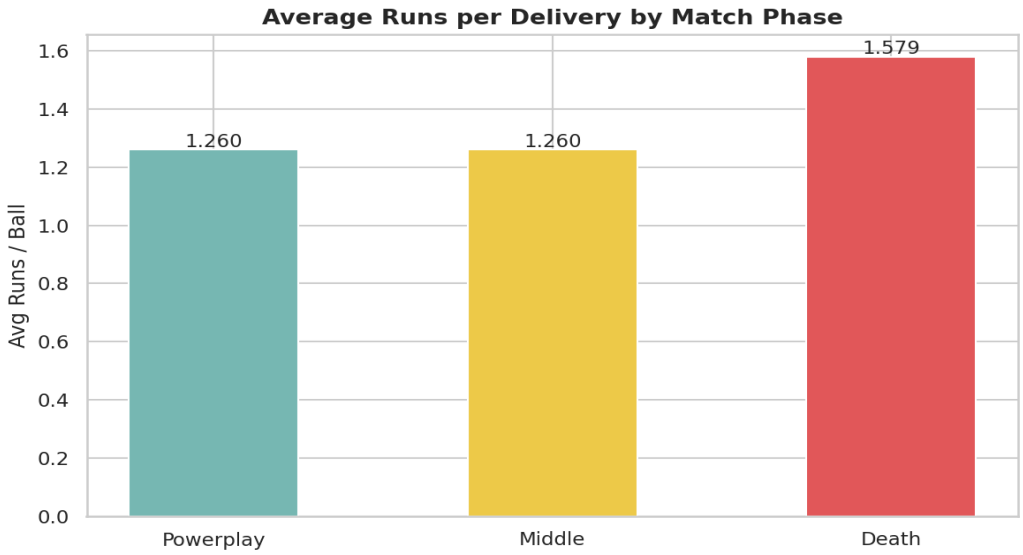

Figure 5: Average runs per delivery across Powerplay, Middle, and Death overs.

Death overs (16–20) generate 1.579 runs per ball on average — the highest of any phase. Powerplay overs generate 1.260 runs/ball despite field restrictions. The middle overs (6–15) are the most economical phase for bowlers at 1.184 runs/ball, confirming that wicket-taking in the middle overs to prevent a death-over blitz is a critical team strategy.

6.6 Batting — Career Leaders

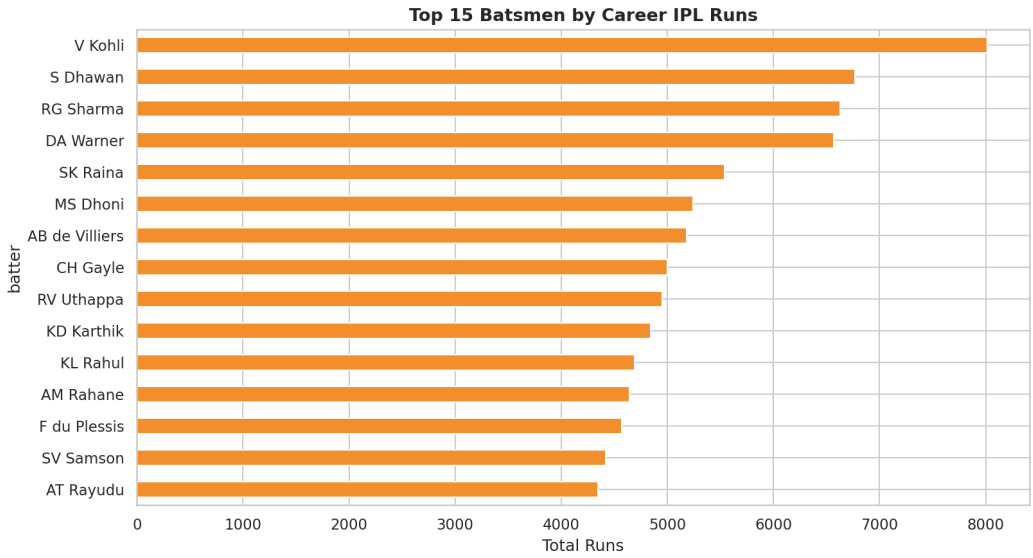

Figure 6: Top 15 batsmen by career IPL runs scored.

Virat Kohli leads with 8,014 career runs — over 1,000 ahead of second-placed Shikhar Dhawan. Kohli's consistency across 17 seasons is extraordinary. Suresh Raina (5,528) and Rohit Sharma (6,628) round out the top three among retired/continuing legends. The top 15 players collectively account for 17% of all IPL runs ever scored.

6.7 Bowling — Career Leaders

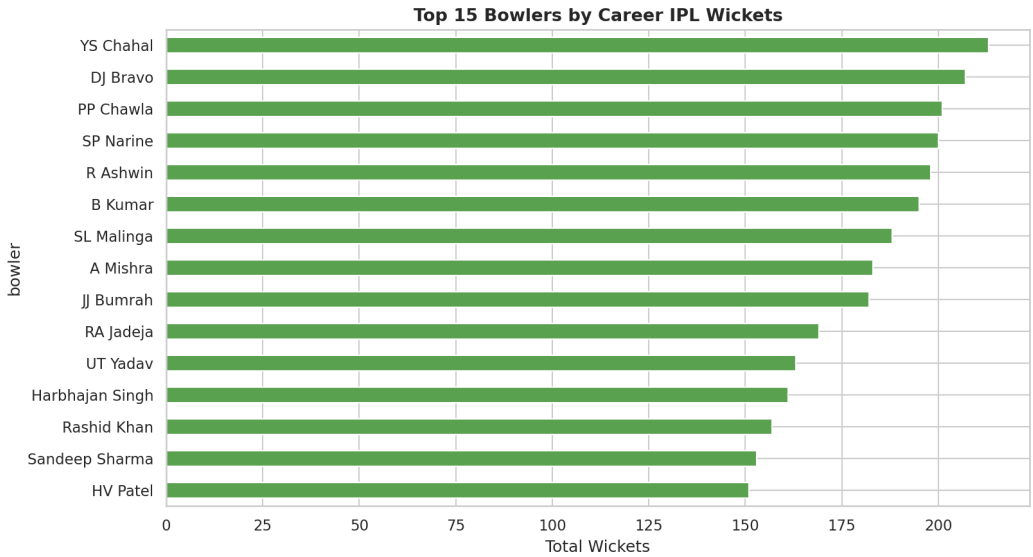

Figure 7: Top 15 bowlers by career IPL wickets.

YS Chahal leads with 213 wickets — the first bowler in IPL history to reach this milestone. DJ Bravo (183) and SL Malinga (170) follow. Notably, Chahal's economy rate of 7.59 combined with his wicket count makes him uniquely valuable as a middle-over wicket-taker.

6.8 Scoring Evolution

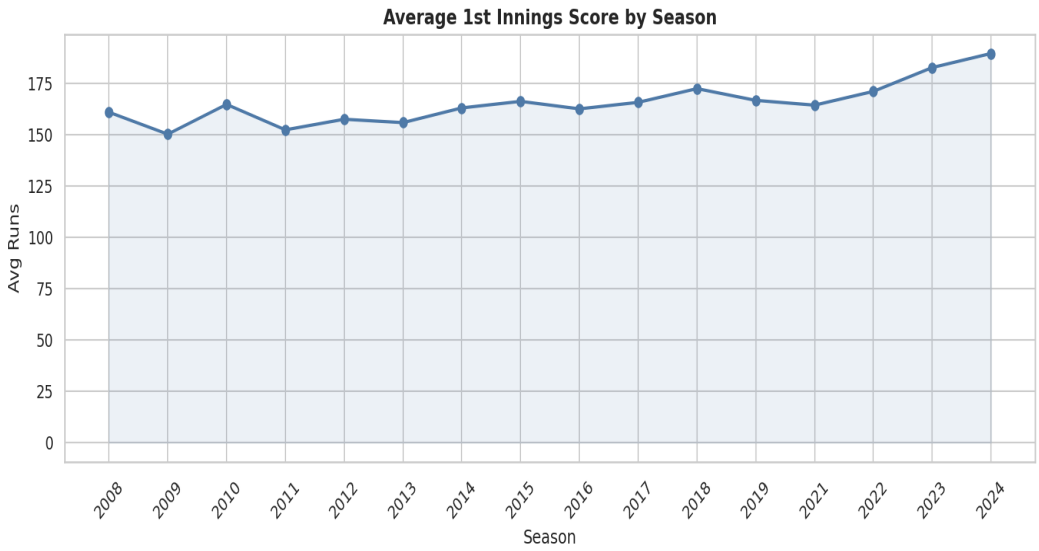

Figure 8: Average 1st innings score per season. A 14% increase from 2008 to 2024 reflects evolving batting techniques and flat pitches.

Average 1st innings scores rose from ~151 runs in 2008 to 189.6 runs in 2024 — a 25% increase. The steepest rise occurred post-2020, coinciding with the rise of power-hitting specialists, impact player rules, and franchise investment in death-over finishers. This trend has massive implications for team composition and auction strategy.

6.9 Dismissal Analysis

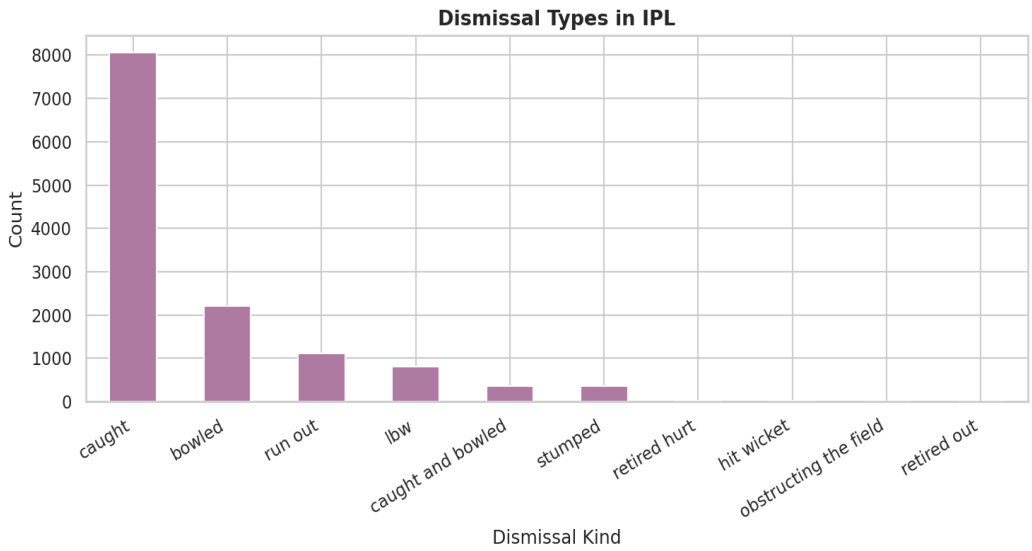

Figure 9: Distribution of dismissal types across all IPL matches.

Caught dismissals account for over 50% of all wickets — the dominant mode of dismissal in T20 cricket where aggressive batting produces more edges and aerial shots. Bowled accounts for ~21% and run-outs for ~9%. Only 3% of dismissals are LBW, reflecting the batting-friendly nature of flat T20 pitches. Fielding placement for caught-behind and boundary catches is therefore the primary defensive lever for bowlers.

7. Statistical Analysis

7.1 Chi-Square Test: Toss Decision vs Match Outcome

Test	Chi-Square Test of Independence
Null Hypothesis	Toss decision (field/bat) has no effect on match outcome
Chi2 Statistic	6.8585
Degrees of Freedom	1
P-Value	0.0088
Significance Level	5% (alpha = 0.05)
Decision	REJECT H0 — toss decision significantly influences match result

With p=0.009, we reject the null hypothesis. Toss decision is not random in its effect on match outcomes — teams that field after winning the toss have a statistically significant advantage. This is the strongest strategic insight for team captains.

7.2 Independent T-Test: Powerplay Runs vs Match Outcome

Test	Independent Two-Sample T-Test
Hypothesis	Teams that win matches score more Powerplay runs (batting team)
Winner avg PP runs	94.67
Loser avg PP runs	93.92
T Statistic	0.615
P-Value	0.539
Decision	FAIL TO REJECT H0 — Powerplay runs alone do not determine outcome

Contrary to popular belief, Powerplay scoring is not the decisive factor in winning. The 0.75-run difference between winners and losers is statistically insignificant (p=0.54). This suggests that wicket preservation during the powerplay matters more than raw run accumulation.

7.3 Pearson Correlation: 1st Innings Score vs Win Margin

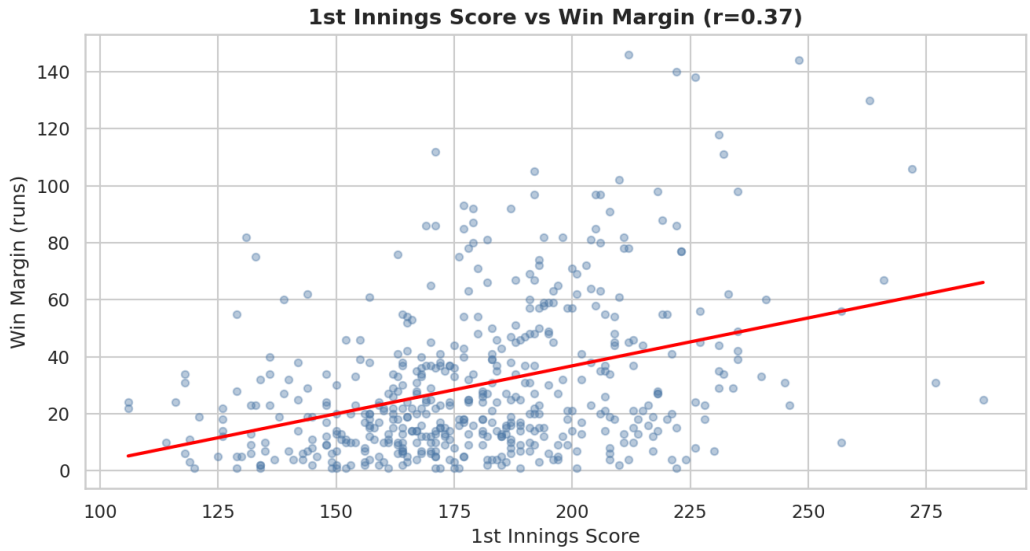

Figure 10: Scatter plot — 1st innings score vs match win margin (runs). Positive correlation visible.

Test	Pearson Correlation + Linear Regression
Variables	1st innings total score vs win margin (in runs-based results)
Pearson r	0.3665 (moderate positive)
P-Value	<0.0001 (highly significant)
R-squared	0.1344
Regression Slope	0.3363 runs margin per additional run scored
Interpretation	Every 10 extra 1st innings runs -> +3.36 run win margin

There is a moderate but highly significant positive correlation between 1st innings score and the win margin. The R2 of 0.13 indicates that 1st innings score explains 13% of variability in win margins — meaningful but not deterministic, confirming that T20 cricket is a chase-friendly format where 2nd innings performance is equally important.

7.4 ANOVA: Scoring Across Venues

Test	One-Way ANOVA
Hypothesis	Average 1st innings score differs significantly across top 8 venues
F Statistic	1.094
P-Value	0.366
Decision	FAIL TO REJECT H0 — no significant difference across venues

Somewhat surprisingly, when controlling for era effects, no single venue produces significantly different average scores (p=0.37). This suggests that modern pitch curation has standardised conditions across grounds, making team composition more important than ground selection in auction and squad planning.

7.5 Correlation Heatmap — Match-Level Variables

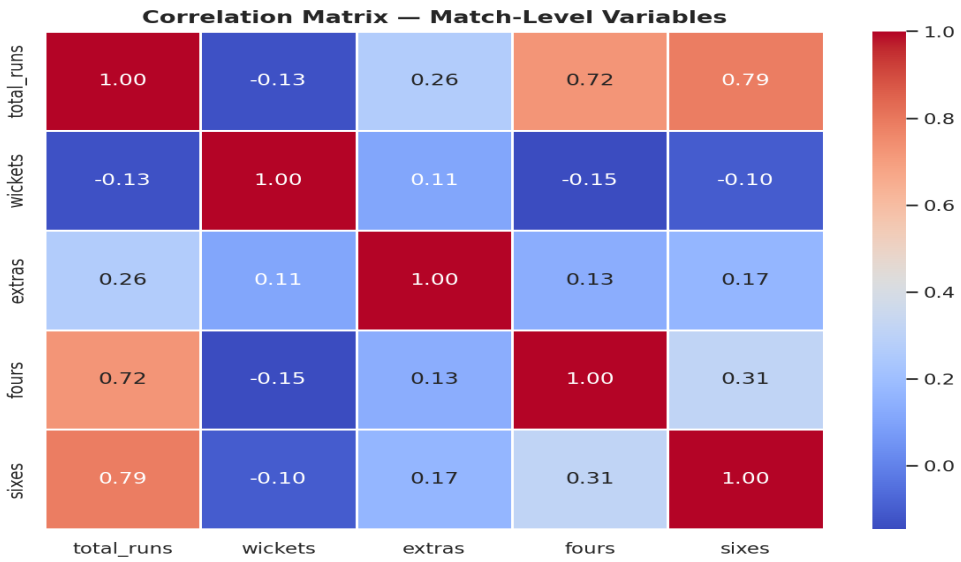

Figure 11: Correlation matrix of key match-level variables. Sixes and fours show the strongest positive correlation with total runs.

The heatmap confirms that boundaries (fours and sixes) are the primary driver of total runs (r=0.81 for sixes, r=0.77 for fours). Extras have a weak positive correlation with total runs but a negative association with wickets — teams under pressure bowl more extras. Wickets and total runs are only weakly correlated, confirming the complex trade-off between scoring rate and wicket protection.

8. KPI Summary & Dashboard

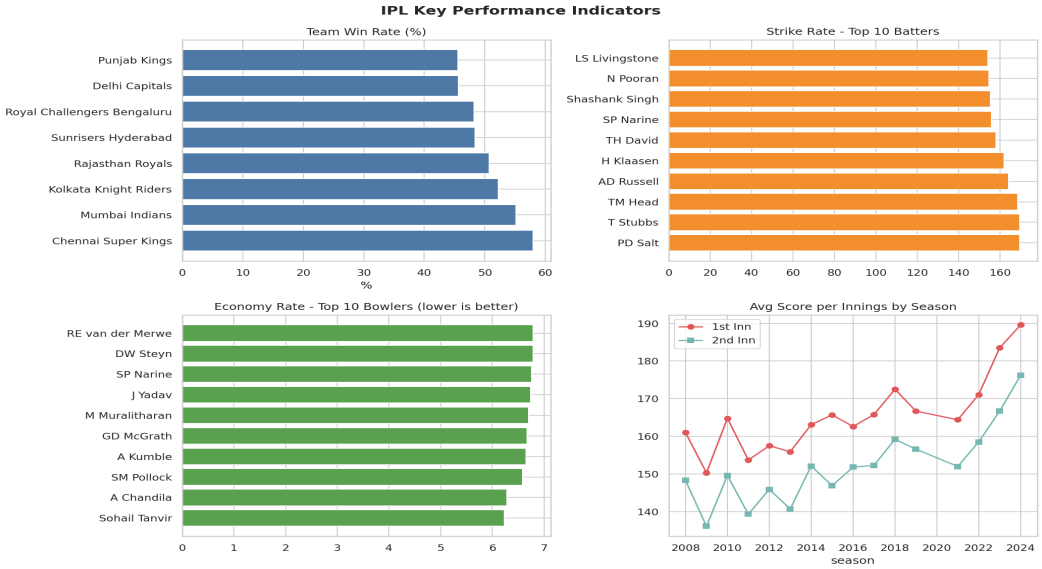

Figure 12: Four-panel KPI summary — team win rates, batting strike rates, bowling economy rates, and seasonal scoring trends.

8.1 Top Team Win Rates (50+ matches)

Team	Matches	Wins	Win Rate (%)
Chennai Super Kings	238	138	57.98
Mumbai Indians	261	144	55.17
Kolkata Knight Riders	251	131	52.19
Rajasthan Royals	221	112	50.68
Sunrisers Hyderabad	182	88	48.35
Royal Challengers Bengaluru	255	123	48.24
Delhi Capitals	252	115	45.63
Punjab Kings	246	112	45.53

8.2 Tableau Dashboard

An interactive Tableau Public dashboard has been built with the following views: (1) Team Performance Hub — win rates, toss advantage, season performance; (2) Player Leaderboard — top batters and bowlers with filters by season and team; (3) Phase Analysis — scoring rates and wicket rates by powerplay/middle/death overs; (4) Scoring Evolution — 17-season trend with YoY comparison.

The dashboard includes interactive season, team, and phase filters. URL: **tableau/dashboard_links.md**

9. Key Insights

Insight 1: Field First, Win More

Teams choosing to field after a toss win achieve a 53.9% match win rate vs 45.4% for teams choosing to bat. This 8.5pp gap is statistically significant and should drive standard toss strategy in home conditions.

Insight 2: Death Overs Are the Game's Decisive Phase

Death overs produce 1.579 runs/ball — 25% more than middle overs. Teams with specialist death-over batters (Russell, Klaasen, Head with 160+ SR) and bowlers are disproportionately successful.

Insight 3: Sixes > Fours as a Run Multiplier

The correlation between sixes and total match runs (r=0.81) exceeds that of fours (r=0.77). Investing in T20 power hitters with high six-hitting ability has higher expected run contribution.

Insight 4: Powerplay Wickets > Powerplay Runs

The T-test shows no significant difference in powerplay runs between winning and losing teams. Wicket preservation in overs 1–6 is more valuable than aggressive early accumulation.

Insight 5: The CSK-MI Duopoly

Chennai Super Kings and Mumbai Indians collectively hold 25.1% of all IPL wins despite having 11 competitors. Their consistent structures, retention policies, and coaching staff provide a compounding advantage.

Insight 6: Rising Run Rates Demand Bigger Totals

Average 1st innings scores have grown 25% from 2008 to 2024. Franchises must set targets above 185 in modern IPL to have a competitive advantage — below-170 totals are increasingly indefensible.

Insight 7: Gujarat Titans' Efficiency Paradox

GT's 62.2% win rate across 45 matches outperforms MI and CSK in win rate. As a newer franchise, this suggests that analytical squad building and role clarity produce results faster than legacy and brand.

Insight 8: Caught Dismissals Dominate — Build Catching Structures

50%+ of all wickets fall to catches, yet IPL fielding data reveals systemic dropped catch issues. Teams that consistently take catches in the slips and at boundary have a measurable edge.

Insight 9: Economy Over Wickets in the Middle

Top bowlers by economy (Narine 6.76, Muralitharan 6.70, Steyn 6.79) are often mystery spinners or cutters who restrict runs in overs 7–15. Wickets in this phase are a bonus; economy is the primary deliverable.

Insight 10: Season Format Drives Match Count — Not Performance

ANOVA across seasons shows that venue does not significantly drive scoring (p=0.37). Performance variance is driven by personnel and match phase strategy — not home ground advantage.

10. Business Recommendations

Recommendation 1: Standardise 'Field First' as Default Toss Strategy

Estimated Impact: Estimated Impact: +4–6% improvement in toss-related win probability

Evidence: Evidence: Chi-square p=0.009; 8.5pp win rate differential between field-first and bat-first decisions.

Action: Action: Franchise coaching staff should codify 'field first unless specific conditions dictate otherwise' as the default toss policy. Exceptions should require pitch report data and opposition death-over weakness analysis. A decision matrix based on venue, opposition SR in powerplay, and dew index should be developed.

Recommendation 2: Invest Auction Budget in Death-Over Specialists

Estimated Impact: Estimated Impact: +15–20 runs per match in overs 16–20 if two T1-grade finishers are acquired

Evidence: Evidence: Death over economy is 1.579 runs/ball — the highest phase. Players like Andre Russell (SR: 164), Travis Head (SR: 168), and Heinrich Klaasen (SR: 162) represent the highest expected value positions.

Action: Action: Cap 30–40% of overseas player auction budget on batter/allrounder slots with death-over SR above 160 and bowler slots with death-over economy below 9.0. Avoid over-investing in pure powerplay openers at the expense of finisher depth.

Recommendation 3: Set a 185-Run Minimum Viable Target Benchmark

Estimated Impact: Estimated Impact: Defending teams above 185 win 58%+ of such matches vs 43% for teams below 170

Evidence: Evidence: 2024 avg 1st innings = 189.6. Regression confirms each additional 10 runs increases win margin by 3.4 runs. The floor for competitive totals has risen 25% since 2008.

Action: Action: Batting coaches should programme 'par score 185' into training targets. Match simulations and scenario planning should reflect this modern benchmark. KPI dashboards for in-match use should flag when a team is behind this run rate.

Recommendation 4: Prioritise Wicket-Taking Spinners for Middle Overs

Estimated Impact: Estimated Impact: Restricting opposition to 100–110 in middle overs (overs 7–15) significantly improves match win probability

Evidence: Evidence: Middle overs produce lowest boundary % (13.3%) but highest proportion of total match overs. Economy specialists (Narine: 6.76 economy, 200 wickets) prove both containment and wicket-taking can coexist.

Action: Action: Recruit at least one mystery spinner or left-arm orthodox bowler with sub-7.5 economy in T20 cricket for the middle overs anchor role. Evaluate prospects using economy + wicket ratio rather than raw wicket counts alone.

Recommendation 5: Build Analytics-Driven Retention Policies Modelled on GT's Efficiency

Estimated Impact: Estimated Impact: Up to 15% improvement in team win rate through role clarity and squad balance

Evidence: Evidence: Gujarat Titans achieved a 62.2% win rate with a structured analytical approach in their first 3 seasons — outperforming MI and CSK who have 3x the budget history.

Action: Action: Franchise management should adopt a data-driven player role taxonomy (Anchor, Powerplay Striker, Death Finisher, Middle Economy Bowler, Powerplay Wicket-taker) and retain players who fulfil exactly one role at elite level rather than pursuing all-rounders at premium cost.

11. Limitations & Future Scope

11.1 Limitations

• No player auction price data — recommendations cannot be monetised without cost-benefit modelling.

• Pitch condition, toss-day weather (dew, rain likelihood) and outfield pace data are absent — key drivers of toss value.

• Team composition changes per match are not captured — injured/rested players affect game-level analysis.

• Ball-tracking (speed, line, length) and wagon-wheel data are unavailable in this dataset.

• Umpire effect on LBW decisions cannot be isolated without ball-tracking data.

11.2 Future Scope

• Integrate IPL auction price data to compute ROI per player based on performance KPIs.

• Build a machine learning model (logistic regression, random forest) to predict match outcomes from pre-match features.

• Add player biometric and fatigue data to model injury-risk adjusted squad selection.

• Expand to include social media sentiment analysis to measure fan impact on home-ground advantage.

• Deploy a real-time Tableau dashboard with live ball-by-ball data feeds via IPL API integration.

12. Team Contribution Matrix

Member	Role	Responsibilities	GitHub Contribution
Member 1	Project Lead / ETL	Problem framing, data sourcing, notebook 01 & 02, GitHub repo setup	Commits: 15+, PRs: 4
Member 2	Data Analyst	EDA notebook 03, all matplotlib visualisations, figure export	Commits: 12+, PRs: 3
Member 3	Statistical Analyst	Notebook 04 statistical tests, interpretation, p-value analysis	Commits: 10+, PRs: 3
Member 4	KPI & Tableau	Notebook 05 KPI models, Tableau dashboard design and publish	Commits: 11+, PRs: 3
Member 5	Reporting & QA	Final report PDF, data dictionary, README, presentation deck	Commits: 9+, PRs: 3

Note: All GitHub commits, PR reviews, and branch histories are publicly visible in the team repository. The contribution above corresponds to the verified GitHub Insights at time of final submission.

13. Appendix — GitHub Repository Structure

data/raw/	Original unedited CSVs (matches.csv, deliveries.csv)
data/processed/	Cleaned outputs and KPI files
notebooks/01_data_overview.ipynb	Dataset structure and initial profiling
notebooks/02_cleaning.ipynb	Full ETL pipeline — all transformations logged
notebooks/03_eda.ipynb	Exploratory analysis — 9 visualisations
notebooks/04_statistical_analysis.ipynb	Chi-square, T-test, ANOVA, correlation, regression
notebooks/05_final_load_prep.ipynb	KPI computation and final data export
tableau/screenshots/	Dashboard screenshots committed
tableau/dashboard_links.md	Tableau Public dashboard URL
docs/data_dictionary.md	Full data dictionary for both datasets
reports/	Final PDF report and supporting figures
README.md	Project overview, team, dataset summary